

Parth Mital

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🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app/

Education & Certifications

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

OCI 2025 Generative AI Professional

Oracle Certified Professional; ID: 103485063OCI25GAIOCP

July 2026

Valid until July 2028

Experience

Concept Marketing

June 2026 – Present

Sole Software Developer & Maintainer (Contract)

Vadodara, India

- Rebuilt a legacy **ASP.NET WebForms + SQL Server** ERP into a **Next.js, TypeScript, Prisma, PostgreSQL** application with **58 API routes**; migrated **114K+** records from **42 source tables**.
- Implemented admin/technician operations for billing, inventory, reports, PDFs, Excel import/export, & daily automation.
- Added CI, **512 tests** across **82 files**, Playwright E2E, secret scans, rate limits, security headers, health checks, & encrypted backups.
- Deployed on **July 1, 2026**; production system used by **60+** daily users, managing **2.8K customers**, **28K service calls**, & **1.1K AMC contracts**.

LearnLine Edustation (IntelGrader)

June 2026 – July 2026

AI Engineer Intern

Remote

- Built a **YOLOv8s** crossout pipeline with **100** annotated sheets, **277** boxes, JSON/CSV manifests, & OCR benchmarks that reduced clutter by **7.11%**.
- Developed an Android ops capture app with **Kotlin, Jetpack Compose, CameraX, OpenCV, TensorFlow Lite, & ONNX Runtime**; trained a **MobileNetV3** quality model on **10K** images, reducing model size by **82.5%**.
- Created PDF ingestion, question-bank, doodle lecture, & static learning-page generators with **PyMuPDF, EasyOCR, Bedrock, Groq, Stable Diffusion, LTX-Video, FFmpeg, Playwright, & axe**.

Projects

🐙 NeuroScan AI: Brain Tumour MRI Analysis System

PyTorch, FastAPI, React, VTK.js

- Benchmarked **21 CNN, transformer, & 3D segmentation architectures**; authored **2 manuscripts** under review.
- Built a **3-stage PyTorch pipeline** on **4,705 pHash-deduplicated MRI scans & 368 BraTS 2020 cases** using **70/15/15 splits**, reaching **99.59% detection, 98.58% classification, & 0.7655 Dice**.
- Engineered a review platform with NIfTI processing, 3D scan review, volumetric analysis, JWT auth, scan libraries, & PDF reports.

🐙 Muse: Self-Hosted Music Streaming & Discovery Platform

Next.js, Fastify, PostgreSQL, FastAPI

- Designed a distributed **4-service architecture** spanning streaming, metadata, async jobs, & hifi proxying.
- Added a **Last.fm/MusicBrainz recommendation engine** with seed expansion, overexposure filters, cached homepage shelves, radio, & **25-track queues**.
- Implemented **57 API routes, 19 Prisma models**, durable job workers, JWT auth, Docker Compose deployment, health/metrics endpoints, CI, & **29 Vitest cases**.

Technical Skills

Languages: TypeScript, Python, JavaScript, Java, C, C++

Backend & Databases: Next.js API Routes, Fastify, FastAPI, Prisma, PostgreSQL, SQLAlchemy, SQLite

AI/ML: PyTorch, TensorFlow Lite, YOLOv8, OpenCV, Computer Vision, OCR, LLM Pipelines

Cloud & Quality: AWS Bedrock, Docker, GitHub Actions, Vercel, Vitest, Playwright, CI/CD, Security Audits

Coursework: DSA, Algorithms, DBMS, Operating Systems, Computer Networks, Compiler Design, AI, NLP, Digital Image Processing